

Macroeconomics II, Lecture XIII: Incomplete Markets in General Equilibrium

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- Recap:
 - ① With incomplete markets, ex-ante homogeneous households will be **ex-post heterogeneous** in terms of $\{c, a, y\} \Rightarrow$ no aggregation into representative agent
 - ② With incomplete markets, consumption dynamics influenced by a **precautionary savings** motive
- Today: we will study the interplay between incomplete markets and the aggregate economy
 - ▶ That is, we will set up a general-equilibrium model

Why general equilibrium? A reminder.

- PE enables us to explore how household consumption-saving decisions respond to income, risk, price changes, credit availability, etc.
- But since prices and income processes are endogenous objects, PE does not permit exploring how economy-wide objects respond to fiscal policy, technical change, labor market shocks, etc.
- Economy-wide objects: the distribution of income and wealth, aggregate savings, aggregate demand, the equity premium, etc.
- First-generation GE models focused on steady state: Bewley (1986), Imrohoroglu (JPE 1989), Huggett (JEDC 1993) and Aiyagari (QJE 1994)
- Second generation added aggregate shocks: Krusell-Smith (JPE 1998)
- Currently, research using business-cycle models with heterogeneous agents is growing rapidly

- ① The static Aiyagari model
 - ① Model setup and definition of stationary recursive equilibrium
 - ② The Aiyagari diagram

- ② Applications
 - ① Can the model explain US wealth inequality?
 - ② Efficiency and fiscal policy

The Aiyagari Model

- The Aiyagari model = neoclassical growth model with a continuum of households that face uninsurable idiosyncratic income risk
- Three ingredients
 - ① Households that solve an income-fluctuations problem
 - ★ Determines supply of labor and assets, demand for consumption
 - ② Competitive firms that maximize production using a production function
 - ★ Determines demand for labor and assets, supply of consumption
 - ③ Market clearing conditions
 - ★ Goods demand = Goods supply
 - ★ Labor demand = Labor supply
 - ★ Asset demand = Asset supply

Household preferences

- A unit mass of households, indexed by i , solve an income-fluctuations problem
- Preferences: households seek to maximize the objective

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_{it})$$

- $U(C)$ satisfies the usual regularity conditions
- Labor supply is exogenous, no disutility from labor

Household income

- Household income: $\epsilon_{it} W_t$
- Household effective labor supply ϵ varies stochastically
- ϵ has finite support $\epsilon \in \mathcal{E} = \{\epsilon_0, \epsilon_1, \dots, \epsilon_N\}$ and follows a **Markov process** with transitions governed by

$$\pi(\epsilon', \epsilon) = \Pr(\epsilon_{t+1} = \epsilon' | \epsilon_t = \epsilon)$$

- The transition probabilities are the same for all households
- This formulation of the income process allows for an arbitrary level of persistence (in contrast to the transitory-permanent income process considered in last lecture)
 - ▶ See Tauchen (EL 1986) and Tauchen-Hussey (Ecmtra 1991) for how to map a continuous AR process into a discrete-state Markov process

Markov processes I

- Let $\mathcal{E} = \{0, 1\}$ (unemployed and employed)
- Collect transition probabilities in matrix T :

$$T = \begin{pmatrix} \pi(0, 0) & \pi(0, 1) \\ \pi(1, 0) & \pi(1, 1) \end{pmatrix}$$

- Suppose t is a quarter. What is the interpretation of $\pi(1, 0)$? $\pi(0, 1)$?
- Let $\pi_{it}(\epsilon)$ be the probability distribution over an individual household i 's state in period t
 - ▶ If we know household i is unemployed in period t , then $\pi_{it}(\epsilon) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- $\pi_{it+s}(\epsilon)$ is given by

$$\pi_{it+s}(\epsilon) = T^s \pi_{it}(\epsilon)$$

- ▶ If we know the household is unemployed in period t , then

$$\pi_{it+1}(\epsilon) = T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \pi(0, 0) \\ \pi(1, 0) \end{pmatrix}$$

- Let $\Pi_t(\epsilon)$ be the distribution of all households across states ϵ in period t . $\Pi_{t+s}(\epsilon)$ is similarly given by

$$\Pi_{t+s}(\epsilon) = T^s \Pi_t(\epsilon)$$

- Stationary distribution $\Pi^*(\epsilon)$ satisfies

$$\Pi^*(\epsilon) = T \Pi^*(\epsilon)$$

- If T is **regular** (all entries of T^n for some $n > 0$ are positive), $\Pi^*(\epsilon)$ is given by

$$\Pi^*(\epsilon) = \lim_{s \rightarrow \infty} T^s \Pi_t(\epsilon)$$

for any starting distribution Π_0 .

- Implication: there is a unique steady state value of aggregate labor supply, given by

$$L = \sum_{\epsilon \in \mathcal{E}} \epsilon \Pi^*(\epsilon)$$

- Households face budget constraint:

$$C_{it} + A_{it+1} \leq \epsilon_{it} W_t + R_t A_{it}$$

- and credit constraint

$$A_{it+1} \geq -\bar{A}$$

- Notes:

- ▶ The savings instrument in this economy is capital, just as in Lectures I-II
- ▶ To make the determination of the real interest rate transparent, we assume here that depreciation happens at the firm
- ▶ R_t is the real gross rate of return on savings in period t

- The representative firm rents labor and capital to produce consumption goods Y using a Cobb-Douglas production function:

$$Y_t = ZF(K_t, L_t) = ZK_t^\alpha L_t^{1-\alpha}$$

- Firm's problem

$$\begin{aligned} & \max_{K_t, L_t} ZF(K_t, L_t) + (1 - \delta)K_t - W_t L_t - R_t K_t \\ = & \max_{K_t, L_t} ZF(K_t, L_t) - W_t L_t - (R_t - (1 - \delta))K_t \\ = & \max_{K_t, L_t} ZF(K_t, L_t) - W_t L_t - (r_t + \delta)K_t \end{aligned}$$

where $r_t = R_t - 1$

- In the optimum:

$$\begin{aligned} r_t + \delta &= F_K(K, L) = \alpha \left(\frac{K_t}{L_t} \right)^{\alpha-1} \\ W_t &= F_L(K, L) = (1 - \alpha) \left(\frac{K_t}{L_t} \right)^{\alpha} \end{aligned}$$

- The goods market clears when

$$\int_{i=0}^1 C_{it} di + K_{t+1} - (1 - \delta)K_t = Y_t$$

- The asset market clears when

$$K_t = \int_i A_{it} di$$

- The labor market clears when

$$L_t = \sum_{\epsilon \in \mathcal{E}} \epsilon \Pi_t(\epsilon)$$

Rational expectations with heterogeneous households

- Note: R_{t+1} and W_{t+1} depend on aggregate savings A_{it+1} , which, via household decision functions, in turn depends on the joint distribution of A_{it}, ϵ_{it}
- Denote the PDF of this distribution with $\gamma(A_{it}, \epsilon_{it})$
- To form rational expectations of R_{t+1}, W_{t+1} , households need to know $\gamma(A_{it}, \epsilon_{it})$
- $\Rightarrow \gamma$ is, in general, a state variable in the household problem
 - ▶ γ is a function, an infinite-dimensional object
 - ▶ This appears bothersome; recall the “curse of dimensionality”
- In a stationary equilibrium, however, γ is constant and $R_t = R$ and $W_t = W$ for all t
- $\Rightarrow \gamma$ is not a state variable assuming that we start and remain in steady state forever

Stationary recursive household problem

- Defining, computing, and illustrating the equilibrium is easier with a recursive setup
- In a stationary environment, the recursive household problem is

$$\begin{aligned} V(A, \epsilon) = \max_{C, A'} \quad & u(C) + \beta \sum_{\epsilon' \in \mathcal{E}} \pi(\epsilon', \epsilon) V(A', \epsilon') \\ \text{s.t.} \quad & C + A' = W\epsilon + (1+r)A \\ & A' \geq -\bar{A} \end{aligned}$$

- r, W constant $\Rightarrow$ solution functions $V(A, \epsilon), C(A, \epsilon), A'(A, \epsilon)$ are time-invariant

Stationary distribution

- A stationary equilibrium features a constant distribution $\gamma(A, \epsilon)$ with support $\mathcal{A} \times \mathcal{E}$
- Define the transition function $Q((A', \epsilon'), (A, \epsilon))$. Q is given by

$$Q((A', \epsilon'), (A, \epsilon)) = I_{A'=A'(A, \epsilon)} \pi(\epsilon', \epsilon)$$

- Importantly, Q is generated by the household policy functions
- γ' is then given by

$$\gamma'(A', \epsilon') = \int_{\mathcal{A} \times \mathcal{E}} Q((A', \epsilon'), (A, \epsilon)) d\gamma$$

where $d\gamma = \gamma(A, \epsilon) dA d\epsilon$

- A stationary distribution γ^* maps itself onto itself:

$$\gamma^* = \int_{\mathcal{A} \times \mathcal{E}} Q((A', \epsilon'), (A, \epsilon)) d\gamma^*$$

Stationary Recursive Competitive Equilibrium

- An SRCE is a value function $V(A, \epsilon)$, policy functions $C(A, \epsilon)$, $A'(A, \epsilon)$, a distribution $\gamma(A, \epsilon)$, aggregate capital and labor demand K, L , and prices W, R , s.t.

① Given W, R : the value function $V(A, \epsilon)$ and policy functions $C(A, \epsilon)$, $A'(A, \epsilon)$ solve the household Bellman equation

② Given W, R : K, L solve the firm problem

③ The markets for capital and labor clear:

$$K = \int_{\mathcal{A} \times \mathcal{E}} A d\gamma$$
$$L = \int_{\mathcal{A} \times \mathcal{E}} \epsilon d\gamma = \sum_{\epsilon \in \mathcal{E}} \epsilon \Pi^*(\epsilon)$$

④ $\gamma(\cdot)$ satisfies

$$\gamma(A', \epsilon') = \int_{\mathcal{A} \times \mathcal{E}} Q((A', \epsilon'), (A, \epsilon)) d\gamma$$

where Q is defined in the previous slide

Does an SRCE exist?

- Firm and household problems are well-defined: solutions exist and policy functions are continuous in R and W
- Labor supply is constant: solving for W that clears the labor market is done by invoking firm optimality given a market-clearing level of capital
- The question boils down to: does there exist an R s.t. the asset market clears?
- Restated: do the asset demand and asset supply curves cross at least once?

Does an SRCE exist?

- To show existence, we will draw the **Aiyagari diagram** in r, K -space (Do on whiteboard)
 - ▶ $r = R - 1$
- Capital demand $K(r)$ is given by firm F.O.C.: $\alpha \left(\frac{K(r)}{L} \right)^{\alpha-1} = r + \delta$
 - ① decreasing
 - ② $r \rightarrow -\delta : K(r) \rightarrow \infty$
 - ③ $r \rightarrow \infty : K(r) \rightarrow 0$
- Capital supply $A(r)$ is given by household savings: $A(r) = \int_{\mathcal{A} \times \mathcal{E}} A'(A, \epsilon; r) d\gamma$
 - ① The stationary state replicates itself forever: $A(r) = \text{long-run mean level of assets}$
 - ② At the lower end, we have that $\lim_{r \rightarrow -1} A(r) = -\bar{A}$
 - ③ What about the upper end?
-

- Without income risk, $\epsilon = 0$, it is straightforward to show that if
 - ① $\beta(1+r) > 1$, consumption and savings grow indefinitely
 - ② $\beta(1+r) = 1$, consumption is constant, savings is bounded
 - ③ $\beta(1+r) < 1$, consumption and savings decrease until the borrowing constraint binds; in the limit, $C = \bar{Y}$
- Why? If credit constraint does not bind, then the household problem is characterized by

$$U_c(C) = \beta(1+r)U_c(C')$$

- Implication: with complete markets (or no income risk), steady state interest rate $R = \frac{1}{\beta}$

Asset convergence: With income risk

- With uninsurable income risk, households have a stronger savings motive
- Implication: assets and consumption grow without bound at lower levels of the interest rate R
- For example, consider the case with $\beta R = 1$
- Optimality conditions

$$\begin{aligned}U_c(C) &= E[V_a(A', \epsilon')] + \mu \\V_a(A', \epsilon') &= U_c(C')\end{aligned}$$

or

$$V_a(A, \epsilon) \geq E[V_a(A', \epsilon')]$$

- Recall: ϵ_N is the maximum of ϵ
- Suppose that the solution is bounded: the maximum of A' is A_{max}
- Evaluating optimality condition at point A_{max}, ϵ_N :

$$\begin{aligned} V_a(A_{max}, \epsilon_N) &\geq E[V_a(A'(A_{max}, \epsilon_N), \epsilon')] \\ &> E[V_a(A_{max}, \epsilon_N)] \text{ using that } V_{aa} < 0 \\ &= V_a(A'_{max}, \epsilon_N) \end{aligned}$$

which is a contradiction. Thus A' does not have an upper bound $\Rightarrow$ asset holdings diverge to infinity

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 - ③ At the upper end, $\lim_{r \rightarrow \frac{1}{\beta} - 1} A(r) = \infty$
- Therefore: asset supply and asset demand cross at least once

Incomplete vs. complete markets

- Consider the steady state of a representative-agent (complete-market) economy
- Euler equation implies:

$$\begin{aligned} u'(C_t) &= \beta(1+r)u'(C_{t+1}) \\ \Rightarrow r &= \frac{1}{\beta} - 1 \end{aligned}$$

- Asset demand is still given by the same firm F.O.C.: $\alpha \left(\frac{K(r)}{L} \right)^{\alpha-1} = r + \delta$
- Result: The addition of uninsurable income risk raises K and depresses r

Stationary state: computational algorithm

- ① Set an initial r_0 that satisfies $-\delta < r_0 < \frac{1}{\beta} - 1$
- ② Solve for K and then W from firm F.O.C.
- ③ Solve household problem given $\{r_0, W\}$, e.g., by value function iteration
 - ▶ This gives you $A'(A, \epsilon; r_0)$
- ④ Solve for γ_T for some large T using the law of motion

$$\gamma_{t+1} = \int_{\mathcal{A} \times \mathcal{E}} Q((A', \epsilon'), (A, \epsilon)) d\gamma_t$$

- ▶ Note: this involves picking an initial distribution γ_0 and then simulating using the computed policy functions

- ⑤ Compute $A_{agg} = \int_{\mathcal{A} \times \mathcal{E}} A'(a, \epsilon; r) d\gamma_T$
- ⑥ If $A_{agg} > K$, set $r_1 < r_0$ and repeat. If $A_{agg} < K$, set $r_1 > r_0$ and repeat until A_{agg} is sufficiently close to K

A note: Aiyagari vs Huggett

- An incomplete-markets model without physical capital is sometimes referred to as a Huggett model, after Huggett (JEDC 1993)
- Example production function:

$$Y_t = F(L_t) = ZL_t^{1-\alpha}$$

- In the case of zero net asset supply (e.g., no government debt, money, foreign borrowing, etc.), asset market clearing is simply

$$\int_{\mathcal{A} \times \mathcal{E}} Ad\gamma = 0$$

- ▶ Zero net asset supply: households can borrow and save in partial equilibrium, but in net, aggregate savings are zero
- The Aiyagari diagram is the same, except that the aggregate demand curve is vertical at zero
 - ▶ See your problem set

Applications

- Now that we have an incomplete-markets stationary equilibrium, let's put it to work
- Questions:
 - ▶ Does the equilibrium wealth distribution match the data?
 - ▶ Can fiscal policy be welfare-improving? Redistributive taxation? Public debt management?

- Taking a realistic income process as given, can the model explain observed wealth inequality?
- This is a theory of wealth inequality entirely based on luck
- Some agents will experience lucky income draws and accumulate assets to insure against bad draws in the future
- Some agents will experience unlucky income draws and deaccumulate assets or borrow to smooth consumption
- Some agents will experience very unlucky income draws and hit the borrowing constraint

US wealth inequality

Quantiles	0	1	5	10	20	40	60	80	90	95	99	100
Earnings	-1,547	0.0	0.0	0.0	0.0	25.7	50.4	87.5	126.1	180.2	497.0	161,523
Income	-506.0	4.2	8.9	12.3	20.1	36.3	58.8	98.7	142.0	207.2	680.7	187,202
Wealth	-474.0	-31.3	-4.6	0.0	7.3	64.7	197.7	496.9	908.4	1,890	8,327	1,411,730

Numbers measured in thousands of dollars. From Díaz-Giménez-Glover-Ríos-Rull (FED Minneapolis 2011). Data from SCF.

	Earnings	Income	Wealth
Coefficient of variation	3.60	4.32	6.02
Variance of the logs	1.29	0.99	4.53
Gini index	0.64	0.58	0.82
Top 1% / lowest 40%	183	88	1,526
Location of mean (%)	69	74	80
Mean / median	1.72	1.77	4.61

From Díaz-Giménez-Glover-Ríos-Rull (FED Minneapolis 2011). Data from SCF.

- Main takeaway: the wealth distribution is much more concentrated than the income distribution

A (standard) parameterization

- $\underline{A} = 0$ or $\underline{A} =$ natural borrowing constraint

- Income process:

- 1 Run Mincer regression

$$\log y_{it} = \alpha + \beta X_{it} + \nu_{it}$$

where X controls for changes in earnings that are predetermined

- 2 Find a process $\mathcal{E} = \{\epsilon_0, \dots, \epsilon_N\}$, $\pi(\epsilon, \epsilon')$ that matches the autocorrelation and variance of ν_{it}

- $\sigma \in [1, 5]$, consistent with experimental evidence on risk aversion

- ▶ $\sigma = 1 \Leftrightarrow \log$ utility

- Calibrated parameters:

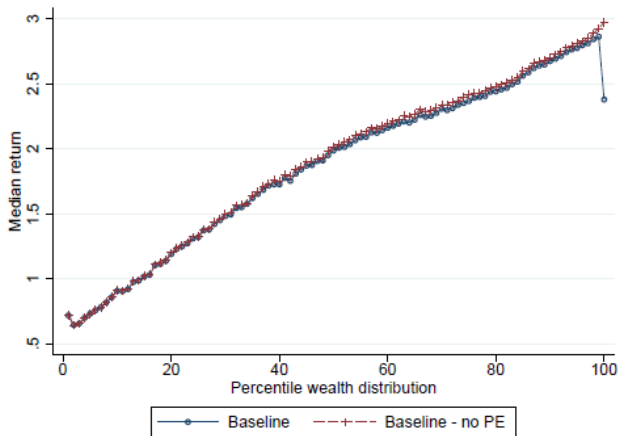
- ▶ α to match labor share $\frac{WL}{Y}$
- ▶ β to match risk-free interest rate r
- ▶ δ to match outside estimates

- Calibrated model typically has wealth Gini coefficient around 0.4
 - ▶ US wealth Gini coefficient is 0.8
- Suggests that precautionary savings can explain some, but not all, of US wealth inequality
 - ▶ It would be very surprising if it could explain everything.
- The failure is due to problems at both ends of the wealth distribution:
 - 1 Very few households are close to the borrowing constraint
 - ★ The welfare loss of not being able to smooth at all is large
 - 2 Very few households have savings several multiples of income at the top of the distribution
 - ★ This is only rational if you face a substantial risk of losing all your income for a long time

What is the model missing?

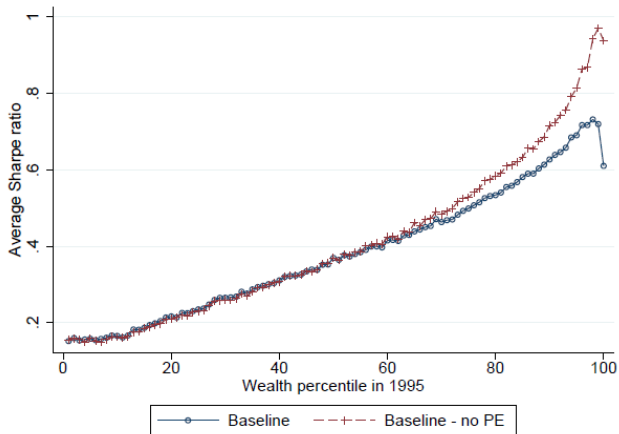
- Suggestions
 - ▶ Adding means-tested social insurance eliminates the saving motive among the income-poor (Skinner-Hubbard-Zeldes, JPE 1995)
 - ▶ Adding bequest savings as a luxury good generates additional savings among the income-rich (De Nardi, ReStud 2004)
 - ▶ β -heterogeneity: given some r , households with low β deaccumulate, households with high β accumulate $\rightarrow$ more wealth heterogeneity within income groups (Krusell-Smith, MD 1997; JPE 1998)
 - ▶ r -heterogeneity:
 - ★ Similar logic to β -heterogeneity
 - ★ In the data, a large fraction of top wealth is held by entrepreneurs. Quadrini (RED 2000) and Cagetti-De Nardi (JPE 2006) model entrepreneurs as households with access to high-return investments
 - ★ See also Krusell-Hubmer-Smith (NBER Macro Annual 2020); Bylund (2026)
 - ▶ Non-zero probability that very rich people lose it all (Díaz-Giménez-Glover-Ríos-Rull, JPE 2003)
 - ★ Authors argue that we miss this in standard calibrations due to top coding in survey income data

Return heterogeneity in Norway



From Fagereng-Guiso-Malacrino-Pistaferri (Ecmtra 2020). Norwegian registry data.

Return heterogeneity in Norway



From Fagereng-Guiso-Malacrino-Pistaferri (Ecmtra 2020). Norwegian registry data.

- Recap: the incomplete-markets equilibrium features a lower r and higher K (and higher Y) than the corresponding complete-markets equilibrium in which $\beta(1+r) = 1$
- Unsurprisingly, this reflects that the market allocation is **inefficient**
- An unconstrained social planner would insure all households against idiosyncratic income risk $\Rightarrow$ constant consumption for all households in the stationary state $\Rightarrow \beta(1+r) = 1$
 - ▶ An unconstrained social planner effectively reintroduces the missing market for a complete set of Arrow securities
- Is the stationary equilibrium allocation **constrained efficient**?
 - ▶ Constrained efficiency: Would a social planner, who can use the same savings instruments as the households, choose different policy functions $A'(A, \epsilon)$, $C(A, \epsilon)$ than households choose in the decentralized equilibrium?
 - ▶ Dávila-Hong-Krusell-Ríos-Rull (Ecmtra 2012): The Aiyagari model is not constrained efficient; the planner wants to instruct well-off households to save more, raising the wage of poor households
 - ▶ $\Rightarrow K(\text{constrained efficient equilibrium}) > K(\text{decentralized equilibrium}) > K(\text{efficient equilibrium})$

- Lack of efficiency opens many doors for welfare-improving policy
- Flodén-Linde (RED 2001) studies redistributive taxation
 - ▶ A flat labor tax τ to finance lump-sum payments T (“basic income”) provides insurance and so raises welfare
 - ▶ At the same time, it depresses incentives to work (labor supply decision easily added to the model)
 - ▶ Flodén and Linde add a labor supply decision to the Aiyagari model and find an optimal level of labor taxes $\tau \approx 0.27$ for a US calibration
- A progressive income tax schedule is an even more effective insurance device
 - ▶ Optimal progressivity is studied by Bénabou (Ecmtra 2002); Conesa-Krueger (RED 2006); Krueger-Ludwig (AER 2013); Heathcote-Storesletten-Violante (QJE 2017)
- Aiyagari-McGrattan (JME 1998) studies government debt management
 - ▶ A higher level of debt B increases the set of liquid assets $\rightarrow$ enables more insurance
 - ▶ Higher B crowds out savings in the capital stock $K \rightarrow$ reduces output and consumption
 - ▶ Aiyagari and McGrattan find that these effects largely offset each other, and the welfare effects from varying the overall debt level are small

What about dynamics?

- So far, our analysis has been static: no aggregate shocks
- Generally, analyzing dynamics requires computational techniques that we do not practice in this course (covered in the first second-year course)
- Central problem: the entire wealth distribution is an infinite-dimensional state variable
- Useful to know: two computational approaches:
 - ① Approximate equilibrium (Krusell-Smith JPE, 1998): exploit that knowing just a few moments of the wealth distribution is sufficient to make very good forecasts of prices
 - ② Linearization (Boppart-Krusell-Mitman, JEDC 2018; Auclert-Bardoczy-Rognlie-Straub, Ecmtra 2021): transition dynamics (IRFs) to a one-time unexpected shock are a sufficient statistic for solving a first-order approximation, just like in representative-agent RBC
 - ★ Computing transition dynamics to a one-time unexpected shock is a bigger system of equations, but conceptually the same as computing the steady state

- Aiyagari model = neoclassical growth model with uninsurable earnings risk
- Key prediction: equilibrium real interest rate lower, due to precautionary savings motive
- Due to incomplete markets, equilibrium is neither efficient nor constrained efficient
- Very general framework for exploring the joint distribution of earnings, consumption, and wealth, and their interaction with macroeconomic dynamics
 - ▶ The literature using incomplete-market models to investigate macroeconomic dynamics has exploded
 - ▶ To conclude the course, I will briefly talk next class about one area of rapidly evolving research: Heterogeneous-Agent New-Keynesian (HANK) models
 - ★ In particular, we will look at a HANK model that also has a frictional labor market, integrating all elements from this course